INCH-POUND

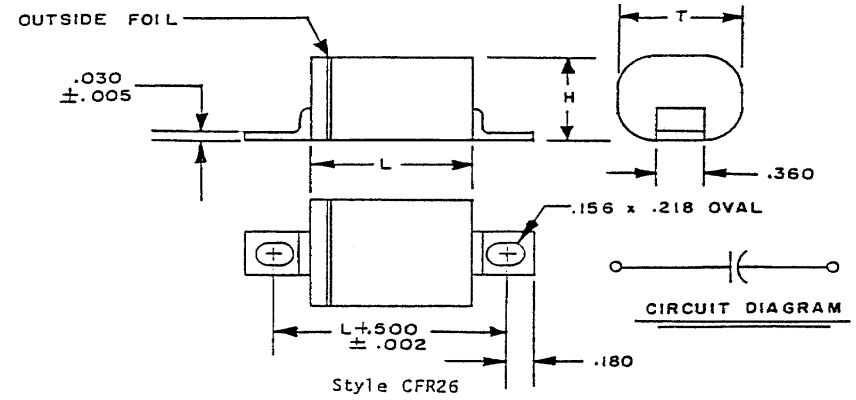
MIL-PRF-55514/11B(CR)
9 October 1997
SUPERSEDING
MIL-C-55514/11A(ER)
14 March 1990

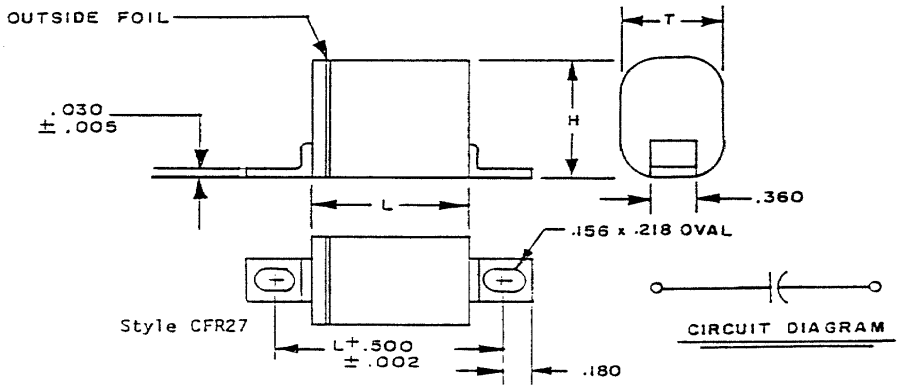
PERFORMANCE SPECIFICATION SHEET

CAPACITORS, FIXED, METALLIZED PLASTIC DIELECTRIC, DC, IN NONMETAL CASES, ESTABLISHED RELIABILITY, STYLES CFR26 AND CFR27

The specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the capacitors described herein shall consist of this specification and the latest issue of MIL-PRF-55514.





Inches	mm
.005	0.13
.030	.076
.046	1.17
.062	1.57
.093	2.36
.156	3.96
.180	4.57
.218	5.54
.360	9.14
.437	11.10
.468	11.89
.500	12.70
.562	14.27
.593	15.06
.718	18.24
.750	19.05
.781	19.84
.812	20.62
.875	22.23
.906	23.01
.937	23.80
.968	24.59
1.000	25.40
1.093	27.76
1.250	31.75
1.281	32.54
1.312	33.32
1.437	36.50
1.687	42.85
1.812	46.02

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
4. These capacitors are to be mounted by screws.

FIGURE 1. Styles CFR26 and CFR27 capacitors.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1.

Case: Non-hermetic.

Dielectric: Metallized polycarbonate.

Characteristic: R.

Capacitance value: See table I.

Capacitance tolerance: See table I.

Solderability: Not applicable.

Operating temperature range: -55°C to +85°C, when properly voltage derated (see figure 2), characteristic "R" capacitors will operate to +125°C.

Voltage rating: See table I.

Preconditioning: Not applicable.

Burn-in: In accordance with MIL-PRF-55514 except for the following conditions:

a. DC burn-in voltage shall be 140 percent of the dc rated voltage applied for 48 hours minimum at 125°C, +4°C, -0°C.

b. AC burn-in: Not applicable.

Insulation resistance: Insulation resistance shall be measured at rated voltage at the applicable temperature. Charge time 2 minutes maximum; however, for capacitance values greater than 1.0 µF, an additional 1 minute per µF is permitted. The following limits shall apply:

Temperature	Insulation resistance
25°C	50,000
85°C	5,000
125°C	500

Marking: In accordance with MIL-PRF-55514.

Qualification submittal:

Type designation	Sample size
CFR27LRA106JM	61
CFR27LRA506JM	61
CFR27LRG506JM	61
CFR27LRB506JM	61

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TABLE I. Capacitor characteristics and dimensions.

Type designation 1/	Capacitance	Capacitance tolerance	Rated voltage at 85°C	ESR milliohms at 100 KhzZ	Ripple current max at 25°C	Resonant frequency kHz	Style CFR26			Style CFR27		
							T 2/	H 2/	L 3/	T 2/	H 2/	L 3/
CFR2-LRA106-- CFR2-LRA206-- CFR2-LRA506--	10.0 µF 20.0 µF 50.0 µF	J, K, M J, K, M J, K, M	50 V 50 V 50 V	11 10 6	12.6 14.8 23.5	265 178 108	.812 .968 1.375	.562 .718 1.000	.750 .875 1.000	.562 .718 1.000	.812 .968 1.375	.750 .875 1.000
CFR2-LRG505-- CFR2-LRG106-- CFR2-LRG206-- CFR2-LRG506--	5.0 µF 10.0 µF 20.0 µF 50.0 µF	J, K, M J, K, M J, K, M J, K, M	75 V 75 V 75 V 75 V	13 12 11 6	11.6 13.5 16.5 24.2	356 251 171 100	--- 1.031 1.312 1.687	--- .718 .937 1.312	--- .875 1.000 1.250	.500 .718 .937 1.312	.750 1.031 1.312 1.687	.875 .875 1.000 1.250
CFR2-LRB505-- CFR2-LRB106-- CFR2-LRB206-- CFR2-LRB506--	5.0 µF 10.0 µF 20.0 µF 50.0 µF	J, K, M J, K, M J, K, M J, K, M	100 V 100 V 100 V 100 V	13 12 11 6	12.4 14.6 17.1 23.6	356 242 160 96	.906 1.093 1.281 1.812	.593 .781 .906 1.437	.875 1.000 1.250 1.375	.593 .781 .906 1.437	.906 1.093 1.281 1.812	.875 1.000 1.250 1.375

1/ The complete type designation will include additional symbols to indicate style, capacitance tolerance, and product level (C, M, P, R, and S).
2/ Tolerances: Thickness and length +.062 inch, -.156 inch (+1.57 mm, -3.96 mm) on .500 inch (12.70 mm) and +.093, -.187 (+2.36 mm, -4.75 mm) on .562 inch (14.27 mm) and greater.
3/ Tolerances: Length +.093 inch, -.062 inch (+2.36 mm, -1.57 mm).

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Substitutability for capacitance tolerance shall be as follows:

Capacitance tolerance	Will replace capacitance tolerance
J	K, M
K	M
M	

Substitutability for voltage shall be as follows:

Volts dc	Will replace voltages
100	75, 50
75	50

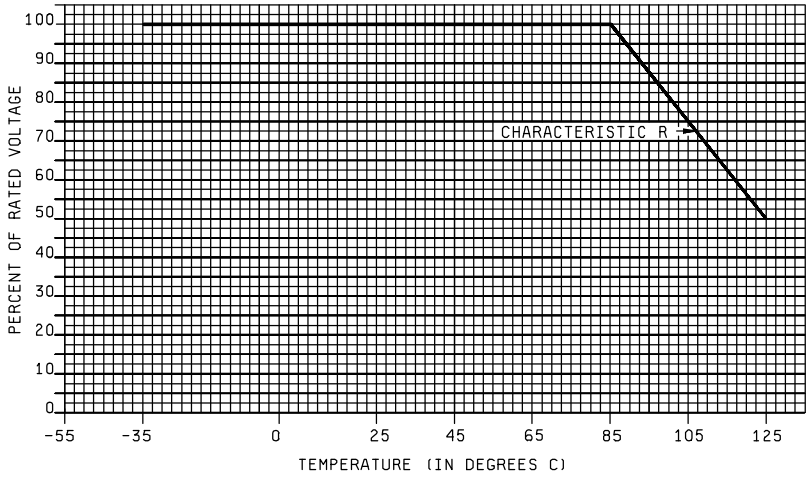

FIGURE 2. Voltage derating with temperature.

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review Activities:
Navy - AS, MC, OS
Air Force - 17, 19, 99
DLA - CC

Preparing activity:
Army - CR

Agent:
DLA - CC

(Project 5910-1960)

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